WARRIOR HACKATHON SCAN - Zero-Trust Android EDR

Version Android License Warrior Hacks

> *An autonomous Android Security Posture Engine that detects, explains, and hardens 10+ critical mobile attack vectors in real-time.*

Built for **Warrior Hacks 2.0 Devpost Competition** by Yousef Esmail Nossier.

🔥 Live Demo

[Live Dashboard Link Here] - Deploy your link

📸 Screenshots

- **Secure State (100/100)**: Zero critical threats, all sandbox policies active
- **Elevated Risk (71/100)**: Detects sideloaded malware & misconfigurations
- **Vulnerability Cards**: CVE-mapped, CVSS-scored, with forensic evidence

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🚨 Problem

80% of Android devices in MENA run sideloaded APKs outside Play Store. Existing scanners only show "virus found". They don't explain *why* a permission is dangerous, *how* Cloak & Dagger hijacks banking apps, or *how* to fix SELinux Permissive kernels.

✅ Solution

Warrior Hackathon Scan is a **Zero-Trust Mobile EDR** with 4 engines:

1. Posture & Vulnerabilities (10 Vectors)

- `CRITICAL CVSS 9.8` - SU Binary & KernelSU/Magisk Root Compromise (`/system/xbin/su`)
- `CRITICAL CVSS 9.3` - Rogue Overlay & Accessibility Hijacking (Cloak & Dagger)
- `HIGH CVSS 8.6` - Sideloaded SharkBot Trojan (High-Entropy Packed DEX)
- `HIGH CVSS 8.2` - SELinux Permissive Mode
- `HIGH CVSS 7.9` - Cleartext HTTP in Network Security Config
- `HIGH CVSS 7.7` - Exported Activity Intent Redirection & Task Hijacking
- `MEDIUM` - USB Debugging (ADB) Left Active in Field - Juice Jacking Risk

2. APK & Package Inspector (5 Samples)

- Static analysis of `AndroidManifest.xml`
- Shannon Entropy Lab: Detects packed/encrypted DEX if entropy > 7.5/8.0
- Signature mismatch vs verified developer keys
- Package installer origin check

3. OS Posture & Kernel

- Android 14 API 34 detection
- Kernel 6.1.43-android14-arm64
- `ro.build.tags` test-keys detection

4. Shannon Entropy Lab [Science]

Scientific malware detection: `Shannon Entropy 7.82/8.0` = High Cryptographic Packing = SharkBot family.

🧬 How It Works - Architecture

```
APK/Device Input
      ↓
```

```
[ Static Parser ] -> AndroidManifest, network_security_config.xml, classes.de
      ↓
[ 4 Analysis Engines ] -> Permissions, Sideloaded Malware, OS Posture, Networ
      ↓
[ Risk Scoring ] -> CVSS 0-10 + OWASP Mobile Top 10 Mapping + Avg CVSS 7.5/10
      ↓
[ Forensic Telemetry ] -> Evidence: "getenforce returns Permissive", "SUID bi
      ↓
[ Autonomous Mitigation ] -> Deploy Defense, Fixed & Patched, Rollback, Audit
```

🛡️ Key Features

- **Real Risk Scoring:** 100/100 SECURE to 0/100 CRITICAL, with Critical/High/Med/Low breakdown
- **Forensic Evidence:** Every finding has `Settings.Global.getInt()`, `entropy=7.82` style proof
- **Autonomous Mitigation:** One-click `Deploy All Sandbox Defenses` & `Apply All Fixes & Hardening`
- **Zero-Trust Compliance:** OWASP Mobile Top 10 - 100% Pass, COMPLIANT status
- **Bilingual UX:** Arabic/English - `تثبيت التطبيق (Install)`
- **Exportable Audit:** Audit Report, Forensic Log, Export Report

🚀 Tech Stack

- Frontend: Vite, TypeScript, Tailwind, Framer Motion
- Analysis: Custom Shannon Entropy calculator, Regex-based Manifest parser
- Security Standards: CVSS v3.1, CVE 2023-40088, CVE-2023-20963, CVE-2022-20005, OWASP MASVS

📦 Installation

```bash
git clone https://github.com/yousefesmail11111-art/warrior-hackson-scan
cd warrior-hackson-scan
bun install
bun run dev
# OR
```

```
npm install && npm run dev
```

Then open `http://localhost:5173` and click **Run Full Android Scan**.

🎬 Devpost Video Script (2:30)

Included in `/DEVPOST.md`

👨‍💻 Creator

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Zero-Trust Mobile Security Framework Active